

Machine Learning

Naïve Bayes- Week 3

Objectives

- Formulation of Naïve Bayes
- Example
- Naïve Bayes on Sparse data
- Pop Quiz

Where to find me?

Office:

Science Hall 250

Office Hours:

**Weds and Thurs 11:00 AM - 1:00 PM EST or by
appointment**

Bayes Rule

Bayes theorem is a fundamental statistical tool used in many scientific disciplines. It is intuitively simple to derive, and states that one conditional probability can be described in terms of the inverse conditional probability scaled by a factor.

Consider two events A and B in some sample space. We have seen that the conditional probability $P(A|B)$ can be written as

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ or, } P(A \cap B) = P(A|B) * P(B)$$

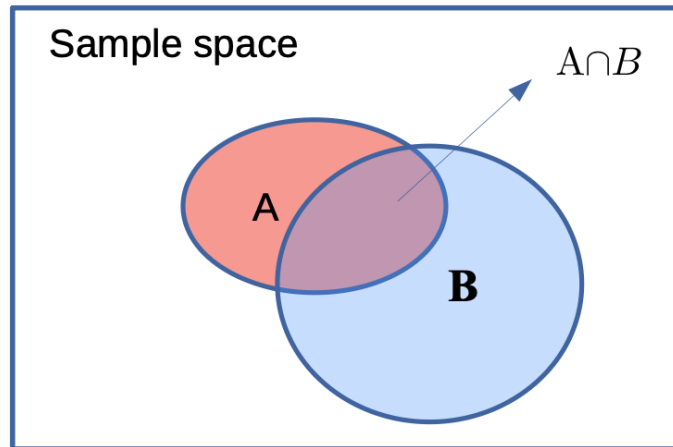
Similarly, $P(B \cap A) = P(B|A) * P(A)$ and since $P(A \cap B) = P(A \cap B)$, we can write:

$$P(A|B) * P(B) = P(B|A) * P(A) \text{ or}$$

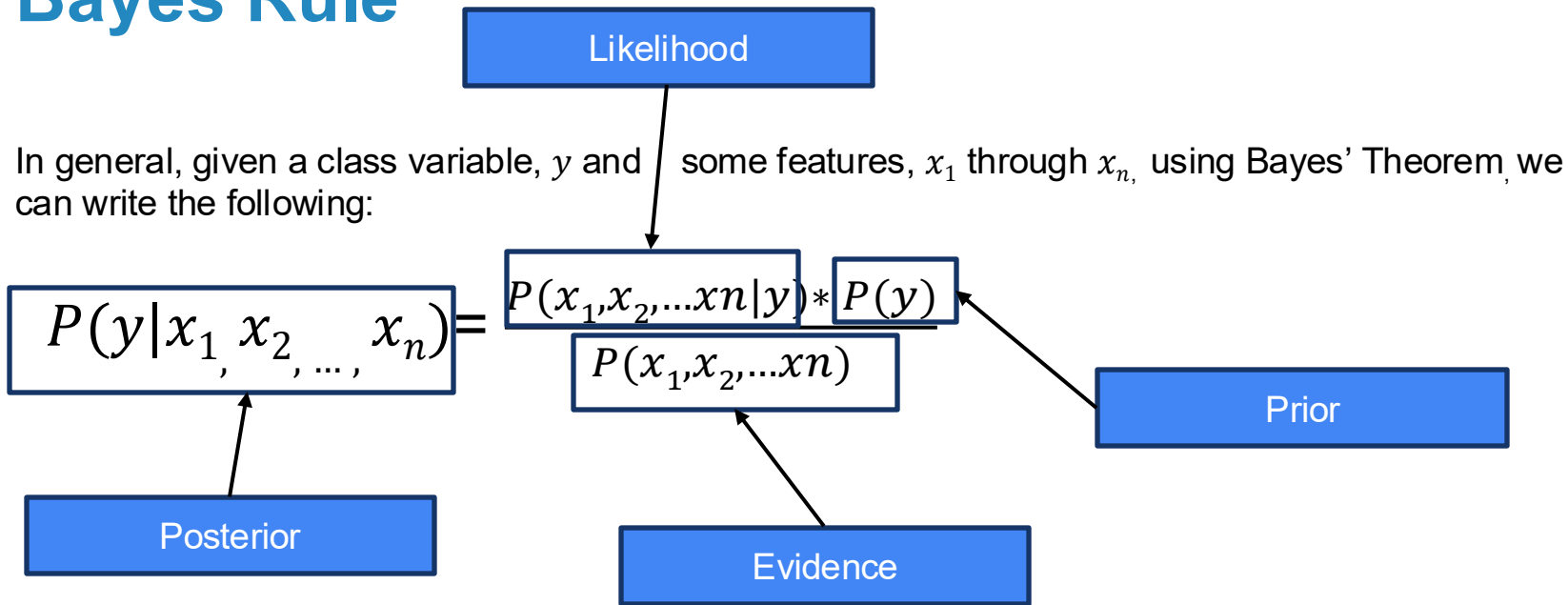
$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$



This is the mathematical statement of Bayes' Theorem



Bayes Rule



In practice, there is **interest** only in the numerator of that fraction, because the denominator does not depend on y and the values of the features are given, so that the denominator is effectively constant.

Naïve Bayes

$$P(y|x_1, x_2, \dots, x_n) = \frac{P(x_1, x_2, \dots, x_n|y) * P(y)}{P(x_1, x_2, \dots, x_n)} \longrightarrow (1)$$

Using the assumption that every feature is independent of every other features , given a specific outcome or category, we can re-formulate (1) as:

$$P(y|x_1, x_2, \dots, x_n) = \frac{P(y) \prod_{i=1}^n P(x_i|y)}{P(x_1, x_2, \dots, x_n)}$$

Therefore, $\hat{y} \propto \operatorname{argmax} P(y) \prod_{i=1}^n P(x_i|y)$

We use Maximum A Posteriori (MAP) estimation to estimate $P(y)$ and $P(x_i|y)$; $P(y)$ is equivalent to the relative frequency of class, y in the training set.

Naïve Bayes Example

Let's look at a fictional weather data:

outlook	temp.	humidity	windy	play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes

outlook	temp.	humidity	windy	play
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Naïve Bayes Example

Frequencies and probabilities for the weather data:

outlook			temperature			humidity			windy			play	
	yes	no		yes	no		yes	no		yes	no	yes	no
sunny	2	3	hot	2	2	high	3	4	false	6	2	9	5
overcast	4	0	mild	4	2	normal	6	1	true	3	3		
rainy	3	2	cool	3	1								
	yes	no		yes	no		yes	no		yes	no	yes	no
sunny	2/9	3/5	hot	2/9	2/5	high	3/9	4/5	false	6/9	2/5	9/14	5/14
overcast	4/9	0/5	mild	4/9	2/5	normal	6/9	1/5	true	3/9	3/5		
rainy	3/9	2/5	cool	3/9	1/5								

Naïve Bayes Example

Now assume that we have to classify the following new instance:

outlook	temp.	humidity	windy	play
sunny	cool	high	true	?

Applying Naïve Bayes to the weather data. Our possible outcomes are {yes, no}. It's often called hypothesis space. Let's denote that as $V = \{yes, no\}$. a_i represents the different features (outlook, temp., humidity, windy)

$$\begin{aligned}
 v_{NB} &= \arg \max_{v_j \in \{yes, no\}} P(v_j) \prod_i P(a_i | v_j) \\
 &= \arg \max_{v_j \in \{yes, no\}} P(v_j) P(\text{outlook} = \text{sunny} | v_j) P(\text{temp} = \text{cool} | v_j) \\
 &\quad P(\text{humidity} = \text{high} | v_j) P(\text{windy} = \text{true} | v_j)
 \end{aligned}$$

Naïve Bayes Example

Computing priors, we have:

$$P(\text{play} = \text{yes}) = 9/14; P(\text{play} = \text{no}) = 5/14$$

Computing conditionals, some examples

$$P(\text{windy} = \text{true} \mid \text{play} = \text{yes}) = 3/9; P(\text{windy} = \text{true} \mid \text{play} = \text{no}) = 3/5$$

Then compute the best class:

$$\begin{aligned} &P(\text{yes})P(\text{sunny}|\text{yes})P(\text{cool}|\text{yes})P(\text{high}|\text{yes})P(\text{true}|\text{yes}) \\ &= 9/14 \cdot 2/9 \cdot 3/9 \cdot 3/9 \cdot 3/9 = 0.0053 \end{aligned}$$

$$\begin{aligned} &P(\text{no})P(\text{sunny}|\text{no})P(\text{cool}|\text{no})P(\text{high}|\text{no})P(\text{true}|\text{no}) \\ &= 5/14 \cdot 3/5 \cdot 1/5 \cdot 4/5 \cdot 3/5 = 0.0206 \end{aligned}$$

Naïve Bayes on Sparse Data

Conditional probabilities can be estimated as relative frequencies:

$$P(a_i | v_j) = n_c / n$$

Where n is the total number of training instances with class v_j and n_c is the number of instances with attribute/ feature a_i and class v_j

This provides a poor estimate if n_c is very small. In extreme cases, n_c can be 0, then the whole posterior is going to be 0.

Solution, we use m -estimate of probabilities: $P(a_i | v_j) = \frac{n_c + mp}{(n + m)}$

Where $m \rightarrow$ equivalent sample size (constant)

$p \rightarrow$ prior estimate of the probability,

Naïve Bayes on Sparse Data

We use m -estimate of probabilities: $P(a_i | v_j) = \frac{n_c + mp}{(n+m)}$

where $m \rightarrow$ equivalent sample size (constant)

$p \rightarrow$ prior estimate of the probability,

In absence of other information, we assume a uniform prior i.e.,

$$p = 1/k$$

Where k is the number of values that the attribute a_i can take.

Summary

- ▶ Bayes classifier combines prior knowledge with observed data: assigns a posterior probability to a class based on its prior probability and its likelihood given the training data.
- ▶ Naive Bayes classifier assumes conditional independence between attributes.
- ▶ Likelihoods can be estimated based on frequencies. Problem: sparse data. Solution: using the m-estimate (adding a constant).

ANY QUESTIONS?